[Aim: 100/100 in Maths]

आरंभ

POLYNOMIALS

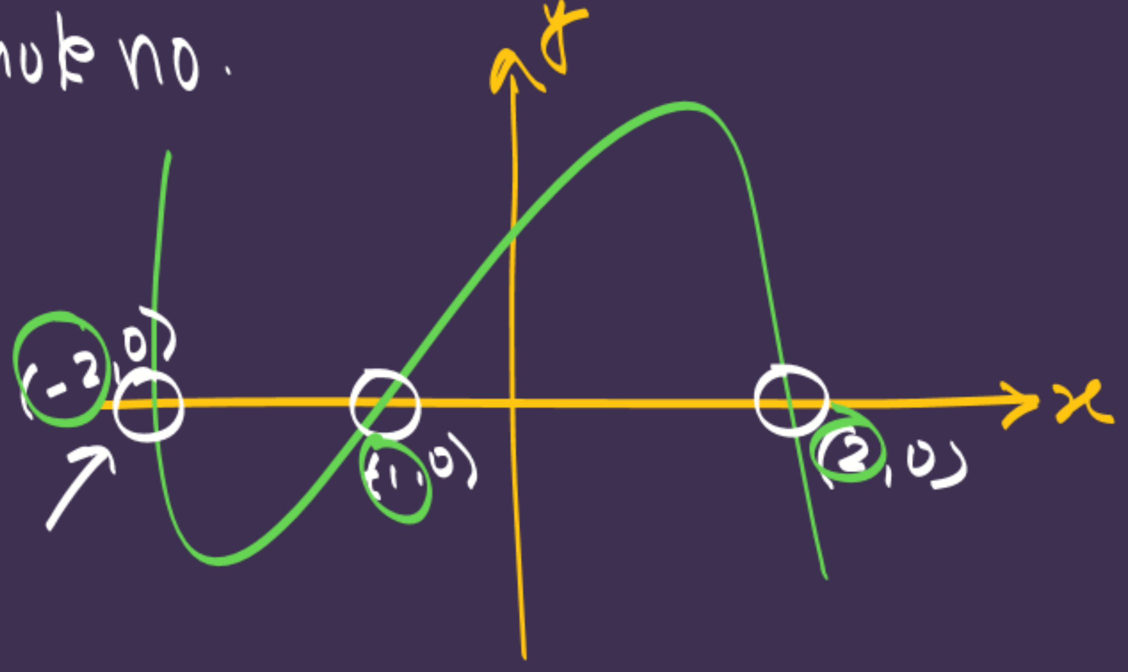
Lecture #4

#ATK²

Polynomial → [variable]$^{\text{whole no.}}$

- no. of terms
  - 1 → monomial
  - 2 → binomial
  - 3 → trinomial
- on degree
  - D = 1 → linear
  - D = 2 → quadratic
  - D = 3 → cubic

K³B → जितनी degree → उतने max. zeroes



no. of zeroes = 3

value of zeroes ⇒ $\{-2, -1, 2\}$

→ Quadratic Polynomial (D = 2)

$$\boxed{ax^2+bx+c}$$

roots → splitting the middle term method

$$\boxed{P(x)=0}$$

#LP:-A rectangular field has area represented by the polynomial $x^2+7x+10$. Which of the following could be its possible dimensions?

(A) $(x+5)(x+2)$ ✓

(B) $(x+7)(x+3)$

(C) $(x+4)(x+1)$

(D) $(x+6)(x+1)$

$p+q=7$

$pq=10$

$$\begin{array}{c|c} 2 & 10 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$x^2+7x+10$$

$$x^2+5x+2x+10$$

$$x(x+5)+2(x+5)$$

$$(x+5)[x+2]$$

$$A = l \times b$$

$$x^2+7x+10 = (x+5)(x+2)$$

product में तोड़ना

$\Downarrow$

factorisation

#LP: p(x) is a polynomial given by:

$$p(x) = -2x + 8x^2 - 1$$

At which of the following points will the graph of p(x) intersect the positive x-axis? **[CBSE CPFQ]**

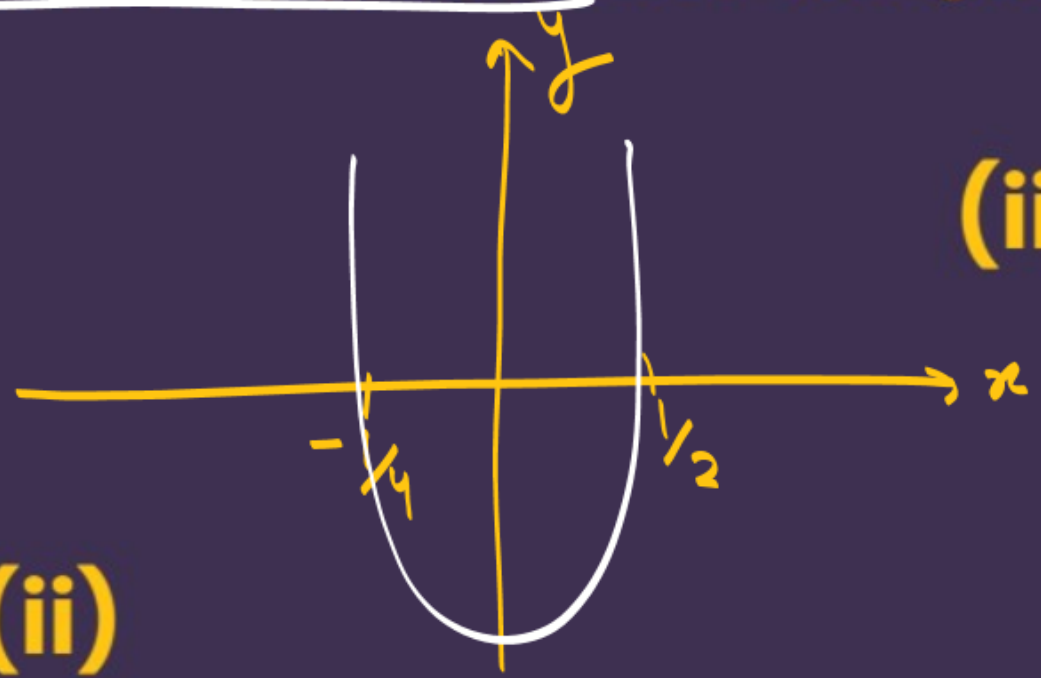
(i) 1/2 (ii) 1/4

(A) only (i)

(B) only (ii)

(C) both (i) and (ii)

(D) none, it never intersects positive x-axis



$p+q = -2$

$pq = -8$

$-2x + 8x^2 - 1 = 0$

+ve

$\boxed{8x^2 - 2x - 1 = 0}$

$8x^2 - 4x + 2x - 1 = 0$

$4x(2x-1) + 1(2x-1) = 0$

$(2x-1)[4x+1] = 0$

$2x - 1 = 0 \quad | \quad 4x + 1 = 0$

$x = 1/2 \quad | \quad x = -1/4$

$\boxed{ax^2 + bx + c}$

$\to a > 0$ ∪ upward parabola

$\to a < 0$ ∩ downward parabola

#LP: Assertion (A): If the graph of a polynomial touches x-axis at only one point, then the polynomial cannot be a quadratic polynomial.

Reason (R): A polynomial of degree n ($n > 1$) can have n zeroes.

[CBSE 2024]

Hw

$P(x)/y$.

#LP: Which of these is the polynomial graphed above?

(A) $(x-2)(x+4) = 0 \longrightarrow 2, -4$

(B) $(x-4)(x+4) \longrightarrow 4, -4$

(C) $\frac{1}{2}(x-2)(x+4) \longrightarrow 2, -4$

(D) $\frac{1}{2}(x-4)(x+2) \longrightarrow 4, -2$

(A) $(x-2)(x+4)$

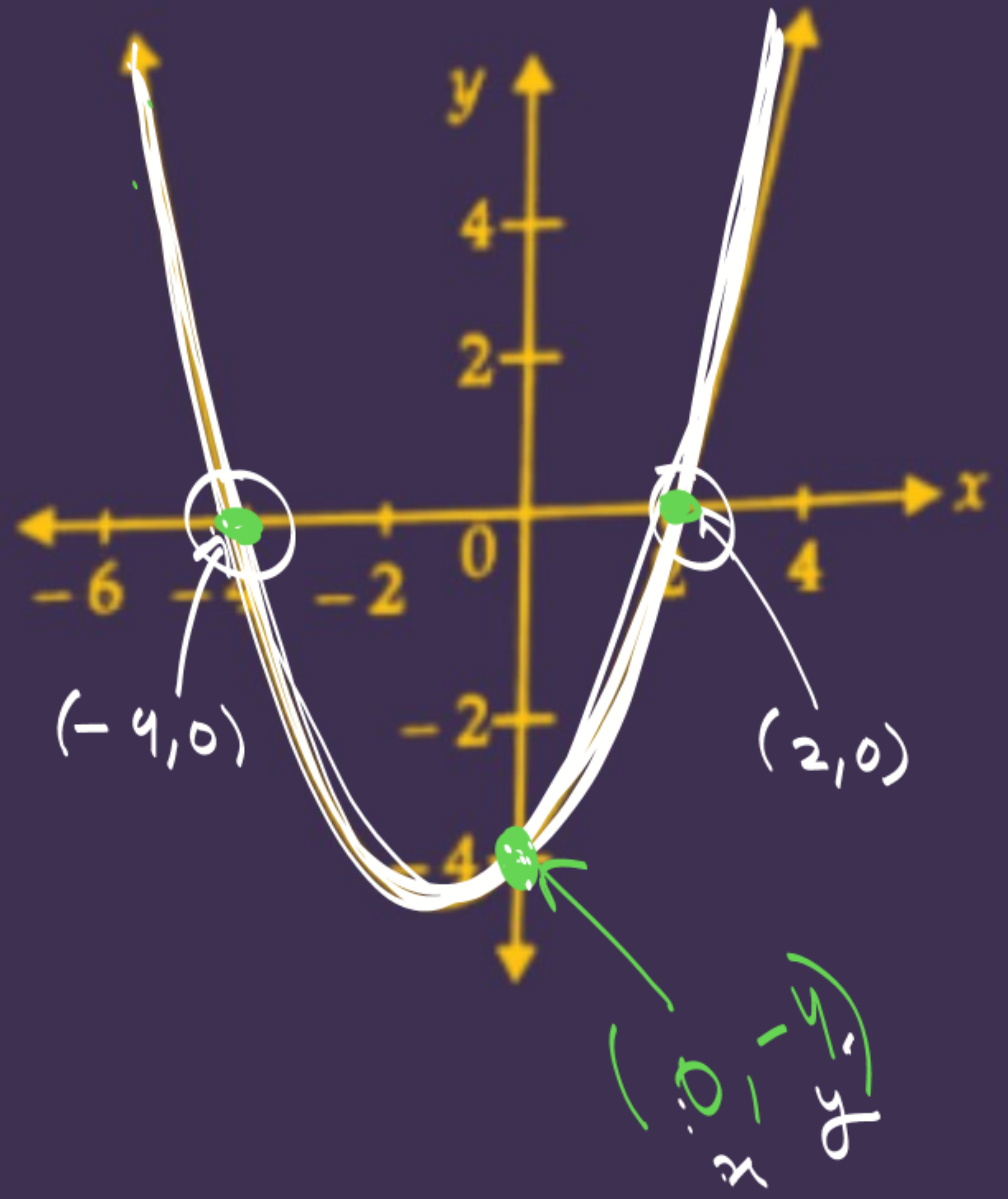
$(0-2)(0+4)$

-8

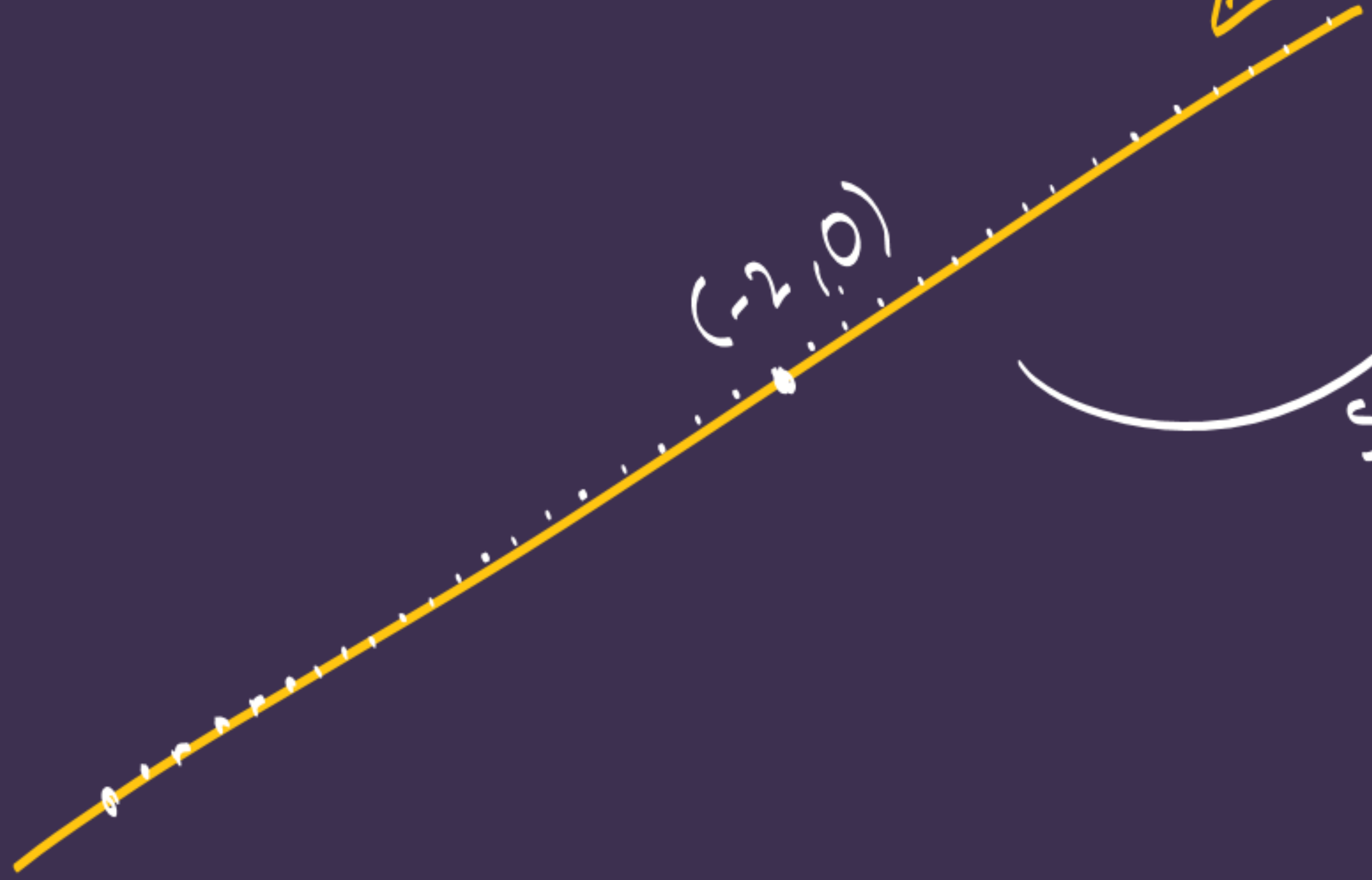
(C) $\frac{1}{2}(x-2)(x+4)$

$\frac{1}{\cancel{2}}(0-\cancel{2})(0+4)$

$y = -4$



$2x + y = 0$



satisfy

Relationship b/w zeroes and coefficients of a Quadratic Polynomial:

$$-2x^2+5x+3 \begin{cases} \alpha \\ \beta \end{cases}$$

$\alpha+\beta = \frac{-b}{a} = \frac{+(5)}{+2} = \frac{5}{2}$

$\alpha \cdot \beta = \frac{c}{a} = \frac{3}{-2} = \frac{-3}{2}$

$$\boxed{ax^2+bx+c} \begin{cases} \alpha \\ \beta \end{cases}$$

Sum of roots (SOR) $\longrightarrow \alpha+\beta = -\frac{b}{a} = -\frac{(\text{coeff of } x)}{\text{coeff of } x^2}$

Product of roots (POR) $\longrightarrow \alpha \cdot \beta = \frac{c}{a} = \frac{\text{constant}}{\text{coeff of } x^2}$

	$3x^2-2x+3$	$-2x^2+2x-3$	ax^2+bx+c $6x+2x^2-3$	$-x^2-x-1$
SOR	$\alpha+\beta=-\frac{b}{a}$ $\Rightarrow -\frac{(-2)}{3}$ $\Rightarrow 2/3$	$\alpha+\beta=-\frac{b}{a}$ $\Rightarrow \frac{\cancel{-}(\cancel{2})}{\cancel{-}\cancel{2}}$ $\Rightarrow 1$	$\boxed{ax^2+bx+c \quad 2x^2+6x-3}$ $\alpha+\beta=-\frac{b}{a}$ $= -\frac{(\cancel{6})^{3}}{\cancel{2}}$ $\Rightarrow -3$	$\alpha+\beta=-\frac{b}{a}$ $=\frac{-(-1)}{-1}$ $\Rightarrow \frac{1}{-1} \Rightarrow -1$
POR	$\alpha\cdot\beta=\frac{c}{a}$ $=\frac{\cancel{3}}{\cancel{3}} \Rightarrow 1$	$\alpha\cdot\beta=\frac{c}{a}$ $=\frac{+3}{+2} \Rightarrow 3/2$	$\alpha\cdot\beta=\frac{c}{a}$ $=\frac{-3}{2}$	$\alpha\cdot\beta=\frac{c}{a}$ $=\frac{\cancel{-}1}{\cancel{-}1} \Rightarrow 1$

#LP: Find the zeroes of each of the following quadratic polynomial and verify the relationship between the zeroes and their coefficients:

(i) $g(s) = \boxed{4s^2 - 4s + 1} = 0$

$\underbrace{4s^2 - 2s}\ \underbrace{- 2s + 1} = 0$

$2s(2s-1) - 1(2s-1) = 0$

$(2s-1)[2s-1] = 0$

$2s-1=0$ | $2s-1=0$

$s = 1/2$ | $s = 1/2$

$\boxed{4s^2 - 4s + 1}$: $1/2$ α, $1/2$ β

$\boxed{\alpha + \beta = \frac{-b}{a}}$

$\frac{1}{2} + \frac{1}{2} = \frac{-(-4)}{4}$

$1 = 1$

LHS = RHS

$\boxed{\alpha \cdot \beta = \frac{c}{a}}$

$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$\frac{1}{4} = \frac{1}{4}$

LHS = RHS

Hence Verified

(ii) $q(x) = \sqrt{3}x^2 + 10x + 7\sqrt{3}$

HW

(iii) $f(v) = v^2 + 4\sqrt{3}v - 15$

HW

(V) $f(x) = x^2 - (\sqrt{3} + 1)x + \sqrt{3}$

HW

HW

#LP: Find the zeroes of the quadratic polynomial $2x^2 - (1 + 2\sqrt{2})x + \sqrt{2}$ and verify the relationship between the zeroes andcoefficients of the polynomial.